

Fact Strategies: Turn It Around, Break It Apart

Answer Key

Grade 3

Quick Answers

- | | |
|--|--------------------------|
| 1. 21 | 6. 54 |
| 2. 24 | 7. product = 40, — |
| 3. 42 | 8. 48 |
| 4. $8 \times 5 = 40$, $8 \times 2 = 16$, $8 \times 7 = 56$ | 9. (a) = 42, (b) = 42, — |
| 5. product = 18, — | 10. product = 63, — |
-

What is verified

6 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.OA.C.7 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 17 tagged checks.

1. Kim knows $3 \times 7 = 21$. What is 7×3 ?

Solution: Turning the factors around never changes the product — a 3-by-7 array turned sideways is a 7-by-3 array with the same 21 things. So $7 \times 3 = 21$. 21

2. $6 \times 4 = 24$, so what is 4×6 ?

Solution: Same two factors, other order: the product stays 24. 24

3. $7 \times 6 = 7 \times 5 + 7 \times 1$

Solution: Six 7s is five 7s plus one more 7: $35 + 7 = 42$. 42

4. $8 \times 7 = 8 \times 5 + 8 \times 2$

Solution: Split the 7 into $5 + 2$. Then $8 \times 5 = 40$ and $8 \times 2 = 16$, and the parts add back together: $40 + 16 = 56$.

$8 \times 7 = 40 + 16 = 56$

5. Turn-around fact for 9×2 , and the product.

Solution: The turn-around fact swaps the factors: $2 \times 9 = 18$. Both facts name the same double: $9 + 9 = 18$. 18

6. $6 \times 9 = 6 \times 10 - 6 \times 1$

Solution: Nine 6s is ten 6s minus one 6: $60 - 6 = 54$. 54

7. Which order is easier to skip count for the 8-and-5 fact?

Solution: Counting by 5s eight times (8×5) is the friendlier count for most students — fives land on tidy numbers. Counting by 8s five times is also correct if that is what the student ran. Either way the product is $8 \times 5 = 40$. 40

8. $4 \times 12 = 4 \times 10 + 4 \times 2$

Solution: Split the 12 into $10 + 2$: $4 \times 10 = 40$ and $4 \times 2 = 8$, so $40 + 8 = 48$. 48

9. Solve 6×7 two ways, then say why they agree.

Solution (a): Break the 7: $6 \times 5 + 6 \times 2 = 30 + 12 = 42$. 42

Solution (b): Break the 6: $7 \times 5 + 7 \times 1 = 35 + 7 = 42$. 42

Solution (c): Both ways count the same 6-by-7 array of 42 things — breaking a factor apart only cuts the array into pieces, it never adds or removes anything, so every correct split must land on the same product.

10. Challenge: any strategy for 9×7 .

Solution: Example strategies: break the 9 into $5 + 4$ ($35 + 28 = 63$); break the 7 into $5 + 2$ ($45 + 18 = 63$); or use ten 7s minus one 7 ($70 - 7 = 63$). Credit any turn-around or break-apart move that is written out and used correctly. The product is $9 \times 7 = 63$. 63